1. Generate message digests and keyed hashes (HMACs) across multiple algorithms (MD5, SHA1, SHA256)

```
root@dan:~# openssl dgst -md5 email.txt
MD5(email.txt)= 0656de7bab59a26431aae7f9f10bac0f
root@dan:~# openssl dgst -sha1 email.txt
SHA1(email.txt)= ceea2a2ae4433223188cea7a874b3fabfd7c8b15
root@dan:~# openssl dgst -sha256 email.txt
SHA2-256(email.txt)= 0f69cf3d6ad1a6d93ff9dcb5c16ca103a157f8b6ee3ad5c4e6da0be972985408
```

I observe that the length of the hashes are different, with the longest one bein the sha256, followed by sha1, and the shortest being md5.

2. **Keyed Hash and HMAC**

```
root@dan:~# openssl dgst -md5 -hmac "abcdefg" email.txt
HMAC-MD5(email.txt)= 7068a9f21d5ef0dfc85678652a0dfcc8
root@dan:~# openssl dgst -sha1 -hmac "abcdefg" email.txt
HMAC-SHA1(email.txt)= 8a79d5cac7be602e65d1fd3f4cb45158a7ec1c8e
root@dan:~# openssl dgst -sha256 -hmac "abcdefg" email.txt
HMAC-SHA2-256(email.txt)= 629484edee21902580c7774f8cdf9073245b5f8910fb48874bf00c8ba59df7ab
root@dan:~# openssl dgst -md5 -hmac "abcdefghijklmnop" email.txt
HMAC-MD5(email.txt)= f67013207099105263074359322 9b3cc
root@dan:~# openssl dgst -sha1 -hmac "abcdefghijklmnop" email.txt
HMAC-SHA1(email.txt)= d616c4b020bd8131767826ad7e76809a3c3e66f8
root@dan:~# openssl dgst -sha256 -hmac "abcdefghijklmnop" email.txt
HMAC-SHA2-256(email.txt)= e0142febcbed24e6d6f968df067ed6f30fef3d8363c47e0b6cf78c2d667b79cd
```

In HMAC the key does not have to be of a fixed size, because it processes the key according to the hash function's block size, not its output size.

```
root@dan:~# openssl dgst -md5 -hmac "IV1013-key" email.txt
HMAC-MD5(email.txt)= 178335489384ee621e569706e059245a
root@dan:~# openssl dgst -sha1 -hmac "IV1013-key" email.txt
HMAC-SHA1(email.txt)= d6fa3d2f9b1912199f5929a7215dec9e24d3b927
root@dan:~# openssl dgst -sha256 -hmac "IV1013-key" email.txt
HMAC-SHA2-256(email.txt)= b0a2a3fb7a088836ae4452696f92912c546d0c1c6e82241b7ad50bd65340ab76
```

3. **The Randomness of One-way Hash**

```
root@dan:/mnt/c/Users/danu1# openssl dgst -md5 email_modif.txt
MD5(email_modif.txt)= d45ffe2a4155e1ba2d4a4e5c6969d063
root@dan:/mnt/c/Users/danu1# openssl dgst -sha1 email_modif.txt
SHA1(email_modif.txt)= 2e66454131fd1580c483f4a29b44acbdea321912
root@dan:/mnt/c/Users/danu1# openssl dgst -sha256 email_modif.txt
SHA2-256(email_modif.txt)= 0c2865ac364a3466bbf7fffdaac3bfdcc960c9bee95dd14578e57300c14562b3
```

I have flipped the first bit of the original input and hashed it. It can be observed that the hash of the modified input is very different from the one of the original, although the size stayed the same.

4. **Strong versus Weak Collision Resistance**

   Weak colision property:

   Original message: IV1013 security
   Matching message: nN2SKsuOfiYuLXrBABE5
   Matching hash: 29688e
   Trials: 22022863

   Original message: Security is fun
   Matching message: 4Nv0QdmJA0a215aFjrpg
   Matching hash: ce91f8
   Trials: 1930474

   Original message: Dan+04/11/2026
   Matching message: 09qpi4DQmu5iniJPCm0H
   Matching hash: f4855f
   Trials: 7556929

   Strong collision property:

   Message 1: PNBgFN0qKYYyAyrCwlyP
   Message 2: UjOLcUJTsYZFGAfs5lTX
   Matching hash: 1a290a
   Trials: 4303

   Based on my observation, strong collision property is easier to break using brute-force than the weak one.